Bachelor's Thesis

Testing for Missingness Patterns in Incomplete Functional Data

Department of Statistics
Ludwig-Maximilians-Universität München

Lukas Stank

Munich, October 23, 2025

Supervised by PD Dr. Fabian Scheipl

Abstract

This thesis offers a comprehensive overview of the statistical tests for assessing the *Missing Completely at Random* (MCAR) assumption in functional data, as introduced by Ofner et al. (2025). It further develops a dedicated implementation of these tests as an extension to the `tidyfun` package (Scheipl et al., 2025) for `R`. In addition to the practical contribution, the thesis primarily explains the theoretical background of functional data analysis and different missingness mechanisms, before presenting the testing framework together with its underlying methodology. The content is focused on mean- and supremum-based tests, bootstrap procedures, as well as the construction of simultaneous confidence bands. In Chapter 3, these methods are evaluated on simulated and real data examples from Ofner et al. (2025), serving as a replication study to demonstrate the validity of the implementation. The results confirm that the methods align with the findings of the original work, thus providing both a deeper understanding of the methodology and an accessible `R` tool for practitioners.

Contents

1 Introduction

In classical statistics, the observation unit is typically well defined. The majority of datasets encountered in practice range from laboratory reports to surveys to economic records or other types of structured data. In such cases, the observed values can be univariate (e.g., age) or multivariate (e.g., a set of economic indicators). Yet in modern statistics, the observations themselves have evolved from single values to entire trajectories over time or space. These may include medical monitoring curves (e.g., ECG and EEG), intraday electricity demand, satellite temperature profiles, or yield curves. When each observation is a function rather than a finite vector, how must our statistical methodology be adapted? And how can estimation, inference, and diagnostics be carried out in the case of partially observed functions?

These questions lie at the foundation of Functional Data Analysis (FDA). At its core, FDA is a statistical framework for analyzing data that appear as curves, surfaces, or other high-dimensional functional objects. Conceptually, we observe random functions $X_i(t)$ with $i = 1, \ldots, n$, modeled as independent realizations of a stochastic process $X(t)$. To make such infinite-dimensional objects tractable, FDA employs techniques such as basis representation, smoothing, and functional principal component analysis (FPCA), which reduce them to interpretable, lower-dimensional summaries (Shang, 2014; Gertheiss et al., 2024; Koner & Staicu, 2023). In its simplest form, FDA assumes one curve per subject, for example a heart rate profile recorded for each individual. A major strength, however, is its flexibility: FDA can be applied to more complex settings, including (i) longitudinal data with multiple curves per subject, (ii) functional time series where each time point corresponds to a full curve, and (iii) spatial or shape data that require geometry-aware methods (Koner & Staicu, 2023; Srivastava & Klassen, 2016).

In reality, data are rarely fully observed, which creates a fundamental challenge: fragmentary observations can distort estimators such as the mean and covariance, thereby introducing bias. Because most standard procedures are designed for complete data, their direct application to incomplete observations may lead to misleading results. Examples include (i) medicine, where wearable sensors may be removed or fail, (ii) environmental monitoring, where stations may go offline, and (iii) economics and finance, where administrative reporting windows or market rules can truncate records. Such gaps are more than a minor nuisance: they can warp covariance matrices, bias principal components and their scores, and ultimately undermine downstream comparisons or predictions.

This leads to a central identification question. Do the missing parts of the curves occur randomly, or is the missingness instead related to other observed covariates, or even dependent on the very variable we aim to study? In multivariate settings, Little (1988) proposed a global test for the Missing Completely At Random (MCAR) assumption by comparing groups of observations that share the same pattern of missing variables. In contrast to the multivariate case, where the number of missing-data patterns is finite, functional data have an uncountably large number of possible observation patterns because of their continuous domain. As a result, a direct extension of Little's test to functional data is not feasible.

This complication has been recently addressed by Ofner et al. (2025), who proposed a functional analogue of testing for independence. In their setting, independence is considered between the data-generating process $X(t)$ and the observation process $O(t)$, which are analyzed separately. The proposed approach builds on the idea of group comparisons, where curves are divided according to their degree of completeness, distinguishing between fully observed and partially observed functions. Intuitively, if the missingness is unrelated to $X(t)$, then fully observed and partially observed curves should yield indistinguishable estimators on the parts of the domain they share. The basic idea, therefore, is to compare mean functions, distributions, or other summary features between these two groups. If their estimates align closely, this supports the MCAR assumption. On the other hand, significant differences between the complete and the incomplete group indicate that the missingness is related to the underlying process. Detecting such discrepancies is crucial, as it highlights situations where standard estimators like the mean or covariance function may no longer be valid unless adjusted.

The first part of this thesis provides an overview of the two central topics. It introduces the basic ideas of Functional Data Analysis and outlines key tools such as functional principal component analysis, which is commonly used for dimension reduction and representation of functional data. In addition, it discusses the challenge of partially observed functions and provides a short introduction to the main types of missingness mechanisms: (i) missing completely at random (MCAR), (ii) missing at random (MAR), and (iii) missing not at random (MNAR).
The second part builds on the work of Ofner et al. (2025), who proposed functional tests for assessing whether the underlying process $X(t)$ is independent of the observation mechanism $O(t)$ (i.e., the missingness process). Rather than developing a new methodology, the focus is on implementing this testing framework in `R` by extending the `tidyfun` package (Scheipl et al., 2025) and utilizing its existing tools for handling functional data. In Chapter 3, the empirical evaluation replicates the simulation studies and applications of Ofner et al. (2025) on the same data sets (heart rate, electricity demand, and temperature curves). The objective is both to confirm the published findings and to show that the implemented methods work as intended in practice.

2 Theory

2.1 Functional Data Analysis

The shift from vector-valued observations to entire functions is not merely a change in data format, but a conceptual step that requires rethinking many statistical foundations. Standard multivariate methods operate in a Euclidean space, where distances, variances, and correlations are defined with respect to a fixed set of coordinates. By contrast, functional data are infinite-dimensional objects. Each observation is a trajectory $X_i(t)$ defined over a domain $t \in I$, yet in practice, such functions are often only partially observed or subject to missing segments. Estimation is therefore no longer straightforward and needs to be adapted from the ground up. This adaptation requires generalizing the fundamental concepts of mean, variance, and the covariance matrix to their functional counterparts: the mean function and the covariance estimator, whose diagonal represents the pointwise variances. In the following, we introduce the basic concepts of FDA, such as mean functions, covariance estimation, and functional principal component analysis, which provide the foundation for the subsequent tests introduced in Section 2.3.

2.1.1 Mean function and estimation

Let $\{X_i\}_{i=1}^n$ be i.i.d. copies of a stochastic process $X(t)$ on $[0, 1]$. Assume that X is integrable, so that $\mathbb{E}[\|X\|] < \infty$ for an appropriate norm $\| \cdot \|$ on the underlying function space. The population mean function $\mu(t)$ is then defined pointwise for each $t \in [0, 1]$ as the expected value of the process,

$$\mu(t) = \mathbb{E}[X(t)].$$

This function represents the average trajectory and serves as the functional analogue of the mean in classical statistics. The sample mean gives a corresponding empirical estimator based on the observed sample

$$\hat{\mu}(t) = \frac{1}{n} \sum_{i=1}^{n} X_i(t),$$

which approximates $\mu(t)$ by averaging the observed trajectories at each time point. Under standard regularity conditions, the law of large numbers implies that the sample mean $\hat{\mu}(t)$ converges in probability to the true mean function, $\hat{\mu}(t) \xrightarrow{p} \mu(t)$, both pointwise and uniformly on $[0, 1]$ as $n \to \infty$.

2.1.2 Covariance function and estimation

Assume that the mean function $\mu(t) = \mathbb{E}[X(t)]$ exists for all $t \in [0, 1]$. The underlying covariance function of X for $s, t \in [0, 1]$ is defined by

$$k(s, t) = \mathrm{Cov}(X(s), X(t)) = E[(X(s) - \mu(s))(X(t) - \mu(t))] \,.$$

It quantifies the linear dependence between the values of the process at times s and t. In the context of functional data, it describes the covariance structure of the trajectories, thus serving a similar role as the covariance matrix in the multivariate case. A corresponding empirical estimator based on the observed sample is given by

$$\hat{k}(s,t) = \frac{1}{n}\sum_{i=1}^{n}\big(X_i(s) - \hat{\mu}(s)\big)\big(X_i(t) - \hat{\mu}(t)\big).$$

2.1.3 Functional Principal Component Analysis (FPCA)

Functional Principal Component Analysis (FPCA) is one of the core methods for reducing the dimensionality of functional data. It generalizes the concept of multivariate PCA to the functional domain. The aim is to describe the variability of the random process $X(t)$ through a small number of orthogonal components that together capture most of its variation, thus enabling both dimension reduction and an interpretable representation of the data.

The starting point of FPCA is the covariance structure of the process. Let $k(s,t) = \mathrm{Cov}(X(s), X(t))$ denote the covariance function, which characterizes the dependence between the process values at time points s and t. According to Mercer's Theorem, the covariance function can be written as

$$k(s,t) = \sum_{j=1}^{\infty} \lambda_j \, \phi_j(s) \, \phi_j(t),$$

where the eigenvalues λ_j are ordered decreasingly and ϕ_j denote the corresponding orthogonal eigenfunctions (Wang et al., 2016). This representation expresses the covariance function as an infinite sum of separable components, each corresponding to an independent direction and magnitude of variation of the underlying process (Hsing & Eubank, 2013, Chapter 9).

Once the eigenfunctions and eigenvalues of the covariance operator are known, the stochastic process $X(t)$ can be expressed as a weighted sum of these orthogonal modes. This representation, known as the *Karhunen–Loève expansion*,

$$X(t) = \mu(t) + \sum_{j=1}^{\infty} \xi_j \phi_j(t),$$

shows that every trajectory can be written as the mean function $\mu(t)$ plus a linear combination of the eigenfunctions $\phi_j(t)$ weighted by uncorrelated random coefficients ξ_j (Hsing & Eubank, 2013, Chapter 9). Each coefficient can equivalently be expressed as $\xi_j = \sqrt{\lambda_j}\, Z_j$, where $Z_j \sim N(0,1)$ are standardized uncorrelated random variables. These coefficients are referred to as *functional principal component (FPC) scores* and measure how strongly each curve deviates from the mean in the direction of a given eigenfunction. They are obtained by projection,

$$\xi_{ij} = \int_0^1 (X_i(t) - \mu(t))\, \phi_j(t)\, dt.$$

Large positive or negative values of ξ_{ij} indicate that the ith curve differs substantially from the mean in the pattern described by $\phi_j(t)$, while scores close to zero imply that it follows the mean shape along that dimension. By construction, the scores are uncorrelated with

$\mathbb{E}[\xi_j] = 0$ and $\mathrm{Var}(\xi_j) = \lambda_j$, so that each eigenvalue λ_j represents the variance explained by its corresponding eigenfunction ϕ_j. The first few components therefore capture the dominant structure of the data, and the proportion $\lambda_j / \sum_{k=1}^{K} \lambda_k$ indicates the share of total variance explained by the jth component.

In practical applications, $k(s,t)$ and $\phi_j(t)$ are replaced by empirical estimates based on an estimated covariance function $\hat{k}(s,t)$ derived from the sample. The eigenvalues and eigenfunctions $(\hat{\lambda}_j, \hat{\phi}_j)$ are then computed numerically, for example by discretizing the integral equation. After this transformation, each functional observation $X_i(t)$ is represented by its mean function $\hat{\mu}(t)$ and a low-dimensional vector of scores $(\hat{\xi}_{i1}, \ldots, \hat{\xi}_{iq})$ associated with the leading eigenfunctions. A common choice for the number of retained components q is based on the *fraction of variance explained* (FVE) criterion. This approach ensures that the selected components capture at least a specified proportion of the data's total variance. These scores form a new, uncorrelated representation of the data that retains most of its information. Subsequent analyses such as regression, clustering, or hypothesis testing are therefore typically conducted in this reduced FPCA space, allowing for a feasible analysis of the otherwise infinite-dimensional data.

2.1.4 Practical representation with `tf/tidyfun`

A convenient way to represent functional data is provided by the `tf/tidyfun` framework in `R` by Scheipl & Goldsmith (2025) and Scheipl et al. (2025). The `tidyfun` package provides a consistent and convenient way to store and work with functional data within the `tidyverse` framework. Its core data type, the `tf` object, is an independent S3 class designed for functional data representation. It provides a table-like interface and supports common `dplyr` and `ggplot2` operations on functional variables. This structure allows for efficient computation, visualization, and aggregation of functional data which will be used throughout this work for the implementation of the proposed methods.

2.2 Types of Missingness

Another central topic in this thesis will be the various patterns of missingness that commonly occur in real-world data, where observations are often incomplete and some entries remain unrecorded or unavailable. Identifying the reason behind missingness is crucial, as incorrect assumptions can lead to invalid statistical inference. More precisely, drawing conclusions from biased estimates results in flawed interpretations, since any inference based on false premises is inherently invalid. In real-world applications, this may entail misguided decisions and potentially harmful consequences. To prevent the faulty application of statistical methods, we provide a comprehensive overview of the various missingness patterns and discuss the specific challenges that arise when applying this framework to functional data.

The terminology of missingness mechanisms has evolved considerably since Rubin's seminal work in 1976 (Rubin, 1976), where the concept of "Missing at Random" (MAR) was first formally defined. In this context, Rubin distinguished between situations where the observation process is unrelated to the underlying data and those where the probability of missingness depends on observed or unobserved variables. The notion of "observed at

random" introduced in the same paper later formed the basis for what is now commonly referred to as "Missing Completely at Random" (MCAR) (Seaman et al., 2013).
Subsequent developments in the literature, particularly the influential monograph by Little & Rubin (2002), established the framework of MCAR, MAR, and "Missing Not at Random" (MNAR), which remains the dominant classification system today. In some literature, MNAR is also referred to as "Not Missing at Random" (NMAR). However, the definitions of these terms have not always been used consistently. As discussed by Heitjan & Basu (1996), authors often employed different definitions, which sometimes led to confusion in both applied and theoretical contexts.
Many authors have attributed different meanings to the same terminology, occasionally even citing Rubin's original work while referring to a different definition (Seaman et al., 2013). Such inconsistencies underscore the importance of stating explicitly which interpretation of the missingness assumptions is adopted in any given analysis.

2.2.1 Formal definitions and Examples

The following definitions and examples follow standard terminology in the missing-data literature (Rubin, 1976; Little & Rubin, 2002; Seaman et al., 2013).

- **Missing Completely at Random (MCAR)**
  The missingness process is unrelated to both observed and unobserved variables. It reflects a mechanism external to the data, such as random data loss due to technical failures, so neither observed nor unobserved values carry information about whether an entry is missing. Under MCAR, the only substantive consequence is reduced efficiency: effective sample size and domain coverage shrink, thereby increasing variance. Note that this reduction in efficiency is a characteristic inherent in all missingness patterns.

  In the classical multivariate setting, with complete data $Y = (Y_{\text{obs}}, Y_{\text{mis}})$, a missingness indicator matrix M, and model parameters ϕ, the MCAR assumption holds if
  $$\forall Y, \phi : \quad p(M \mid Y, \phi) = p(M \mid \phi),$$
  i.e., $M \perp\!\!\!\perp Y$.

  In functional data, this relation translates to a stochastic process $X(t)$ observed through an indicator process $O(t) \in \{0, 1\}$ (Ofner et al., 2025). The corresponding MCAR condition is then given by
  $$\forall t \in [0, 1], \phi : \quad \mathbb{P}\{O(t) = 1 \mid X(\cdot), \phi\} = \mathbb{P}\{O(t) = 1 \mid \phi\}.$$

  It is important to note that the MCAR assumption does not imply that the occurrence of missingness is itself random. Rather, it states that the probability of missingness is unrelated to both observed and unobserved variables. Under MCAR, estimators of $\mu(t)$ and $k(s, t)$ are unbiased on any observed subdomain.

  Examples:

  - A mail survey on income, where some questionnaires are accidentally lost $\rightarrow$ missingness is unrelated to true income.

  - Heart-rate monitoring study where some participants temporarily remove the device for reasons unrelated to their heart rate (e.g., going to the bathroom) $\rightarrow$ missingness is independent of the heart rate trajectory.

- **Missing at Random (MAR)**
  After conditioning on observed information, the missingness does not depend on the unobserved part of the data. In other words, once we account for all available variables, the probability of missingness is fully explained and no additional dependence on unobserved variables remains. This mechanism is less restrictive than MCAR and often more realistic in practice.

  Given the partition $Y = (Y_{\text{obs}}, Y_{\text{mis}})$, a missingness indicator matrix M, and unknown parameters ϕ, MAR holds if

  $$\forall\, Y_{\text{mis}}, \phi : \quad p(M \mid Y, \phi) = p(M \mid Y_{\text{obs}}, \phi),$$

  i.e., $M \perp\!\!\!\perp Y_{\text{mis}} \mid Y_{\text{obs}}$.

  In the context of functional data, the probability of observing a curve at time t may depend on already observed segments of the trajectory, but not on the unobserved ones (Ofner et al., 2025). Given the partition $X(\cdot) = (X_{\text{obs}}(\cdot),\, X_{\text{mis}}(\cdot))$, MAR is defined as

  $$\forall\, t \in [0, 1], \phi : \quad \mathbb{P}\{O(t) = 1 \mid X(\cdot), \phi\} = \mathbb{P}\{O(t) = 1 \mid X_{\text{obs}}(\cdot), \phi\}.$$

  Under MAR, unbiased estimation is possible if the observed information that drives missingness is properly included in the analysis model. Otherwise, estimators may become biased.

  Examples:

  - A survey on income in which younger participants respond less frequently $\rightarrow$ missingness depends only on the observed covariate, namely age.
  - In a heart-rate monitoring study, segments recorded later in the day are more likely to be missing because participants tend to remove the device in the evening, for instance, when going to bed or taking a shower. $\rightarrow$ Missingness depends on the observed covariate "time", not on the unobserved heart-rate values.

- **Missing Not at Random (MNAR)**
  Even after conditioning on the observed part of the data, the missingness still depends on a latent or unmeasured variable. This is the most problematic mechanism, since ignoring the missingness process typically leads to biased inference and inconsistent estimators unless the mechanism itself is explicitly modeled.

  Formally, the MAR condition fails if:

  $$\exists\, Y_{\text{mis}}, \phi : \quad p(M \mid Y, \phi) \neq p(M \mid Y_{\text{obs}}, \phi),$$

  equivalently, $M \not\perp\!\!\!\perp Y_{\text{mis}} \mid Y_{\text{obs}}$.

Given functional data, this means that the independence between the missingness process and the underlying data no longer holds. More precisely, there exists at least one time point t such that

$$\exists\, t \in [0,1], \phi : \quad \mathbb{P}\{O(t) = 1 \mid X(\cdot), \phi\} \neq \mathbb{P}\{O(t) = 1 \mid X_{\text{obs}}(\cdot), \phi\},$$

i.e., the probability of being observed at time t depends on unobserved parts $X_{\text{mis}}(\cdot)$ or on an unobserved covariate (Ofner et al., 2025).

Examples:

- Individuals with very high or very low income are less likely to report income $\rightarrow$ missingness depends directly on the variable of interest, namely income.
- A temperature sensor systematically fails when the true temperature exceeds a critical value $\rightarrow$ missingness depends on the underlying temperature trajectory.

2.3 Proposed Testing Framework

As addressed in the introduction, the direct application of Little's proposed testing framework is not feasible in an FDA setting. The fundamental difference is that, in the multivariate case, there are only finitely many observation patterns (at most 2^n for n variables), each corresponding to a distinct configuration of observed and missing entries. This allows the data to be partitioned into groups according to these patterns, which form the basis of the test. In contrast, FDA involves uncountably many observation patterns in the form of curves and functions. Ofner et al. (2025) extend the framework of Little to the functional setting, enabling functional data to be grouped in a similar manner.

Example: multivariate setting
Suppose we observe three variables Y_1, Y_2, Y_3. Each individual may exhibit a different observation pattern of observed (1) and missing (0) entries:

ID	Y_1	Y_2	Y_3	Pattern
1	1	1	1	complete
2	1	0	1	missing Y_2
3	1	0	0	missing Y_2, Y_3

Table 1: Observation patterns in multivariate MCAR test (Little, 1988).

Here, the test groups individuals by their observation pattern and compares mean values across the groups. With $n = 3$ variables, there are at most $2^3 = 8$ distinct patterns.

Example: functional data setting
In FDA, each subject provides a trajectory $X_i(t)$ observed only on a subset of the domain. Instead of finitely many missing-value configurations, each curve may be observed on arbitrary intervals. Ofner et al. (2025) therefore suggest partitions such as "completely observed" vs. "partially observed" functions; see Table 2:

ID	Observation domain	Group
1	$[0, 1]$	complete
2	$[0, 0.7]$	incomplete
3	$[0.2, 0.8]$	incomplete

Table 2: Partition of functions according to observation domains.

After grouping, we compare selected estimators, such as the mean, to determine if they differ significantly across the two groups. Note that alternative partitions are also possible, for example, by distinguishing curves that are ”more complete” vs. ”less complete”. In this case, a threshold parameter $\delta \in (0, 1)$ determines how much of the domain a curve must cover to be assigned to the ”more complete” group; for further details, we refer to Section 2.3.2.

2.3.1 Hypothesis

To precisely define our Hypothesis and test setting, we need to establish uniform notation for our observations and grouping. Let $X = (X(t) : t \in [0, 1])$ denote the underlying stochastic process representing the functional data, and let $O : [0, 1] \to \{0, 1\}$ be the indicator process given by

$$O(t) = \begin{cases} 1 & \text{if } X \text{ is observed at time } t, \\ 0 & \text{else.} \end{cases}$$

In practice, we do not directly observe X, but only the product XO. This distinction is crucial: the information in O allows us to distinguish the case when X is truly zero and observed ($X(t) = 0$, $O(t) = 1$) from the case where X is unobserved ($O(t) = 0$). An Illustration of X and O can be seen in Figure 1. For more detailed definitions, we refer to Sections 2 and 2.1 (pp. 3–4) of Ofner et al. (2025).
Based on this framework, we test the *missing completely at random* (MCAR) assumption, which states that

$$H_0 : X \text{ and } O \text{ are independent.}$$

Meaning that the occurrence of missing values is unrelated to the underlying process X. In other words, the probability of observing a curve at a given time t does not depend on the value of $X(t)$. This independence is crucial for many functional data methods, since it ensures that estimators are unbiased.

2.3.2 Grouping

In order to test the MCAR assumption, the functional data need to be divided into groups based on their observation domains. We generally denote

$$A = \text{“complete” functions}, \qquad B = \text{“incomplete” functions.}$$

Following Ofner et al. (2025), the two approaches are defined as follows. For a function $f \in D[0, 1]$, where $D[0, 1]$ is the Skorokhod space of càdlàg functions, let $m(f)$ denote the Lebesgue measure of $\{t \in [0, 1] \mid f(t) = 1\}$:

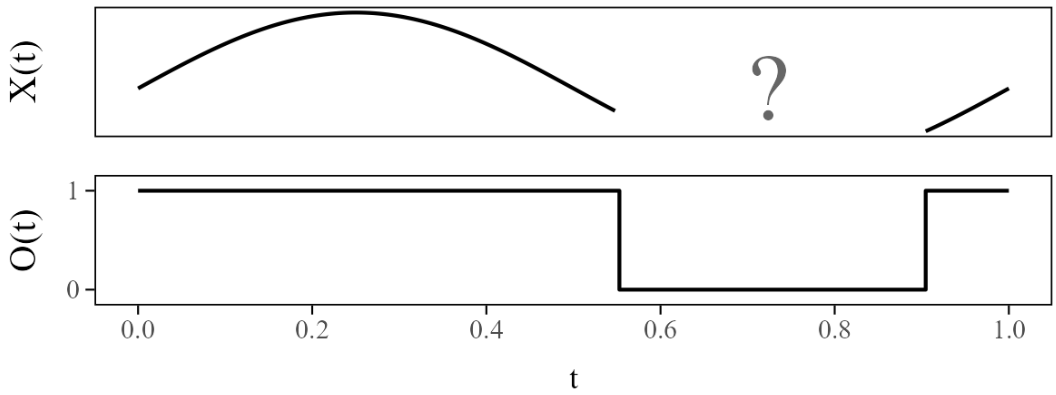

Figure 1: Illustration of random function X and indicator process O.

- **Partition 1: Complete vs. incomplete curves**

$$A = \{f \in D[0,1] : m(f) = 1\} \qquad B = D[0,1] \setminus A$$

Interpretation: Group A contains curves that are completely observed over the full domain $[0,1]$, while B contains all curves that are incomplete at least at some part of the domain. Illustration on the left side of Figure 2.

- **Partition 2: Partition by proportion of observed domain**
For a threshold $\delta \in (0,1)$,

$$A = \{f \in D[0,1] : m(f) \geq \delta\}, \qquad B = \{f \in D[0,1] : m(f) < \delta\}$$

Interpretation: Curves that cover at least a fraction δ of the domain are assigned to A, while curves with less coverage are assigned to B. This is useful when no fully observed curves exist. Illustration on the right side of Figure 2.

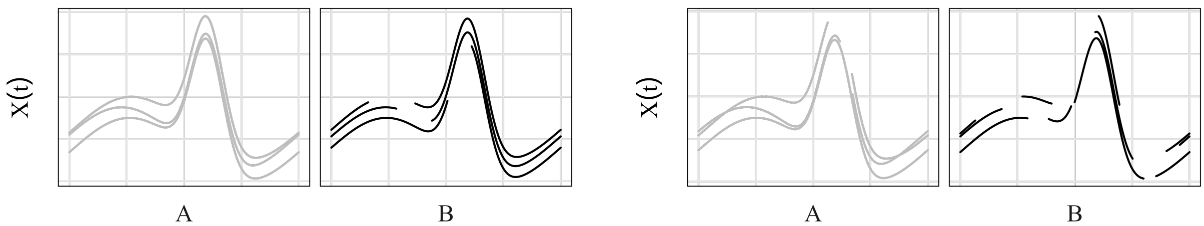

Figure 2: Complete vs. incomplete curves (left); Partition by proportion of observed domain with threshold $\delta = 0.9$ (right).

To ensure the comparison is conducted on a common subdomain, the analysis is restricted to a subset $I \subseteq [0,1]$. This guarantees that, at each time point, a sufficient proportion of curves from both groups is observed. More precisely, the subdomain is defined such that at least 10% of all curves are present in both groups:

$$I = \left\{ t \in [0,1] \;\middle|\; \min\left\{ \sum_{i=1}^{n} O_i(t) \mathbb{1}\{O_i \in A\}, \sum_{i=1}^{n} O_i(t) \mathbb{1}\{O_i \in B\} \right\} > n \times 0.1 \right\}.$$

2.3.3 Mean Comparison via L^2 Test

In the following, we outline concrete testing procedures based on the general framework introduced above. Since the general MCAR hypothesis $H_0 : X \perp\!\!\!\perp O$ is not consistent against the general alternative $H_A : X \not\perp\!\!\!\perp O$, we consider a weaker, testable version formulated in terms of conditional mean functions. Specifically, we test for equality of mean functions across groups A and B:

$$H_0^\mu : \mathbb{E}[X(t) \mid A] = \mathbb{E}[X(t) \mid B], \qquad \forall\, t \in [0,1],$$

which represents a relaxation of the full independence assumption H_0 (i.e., $H_0^\mu \supset H_0$). The reasons why such an adaptation of the null hypothesis is necessary are outlined in Section 4.

Step-by-step outline of the asymptotic-mean L^2 test; see Algorithm 1:

- Step 1: As established earlier, each trajectory is accompanied by an indicator process $O_i(t) \in \{0,1\}$ that specifies whether $X_i(t)$ is observed at time t. Building on this framework, the general setting from Section 2.1.1 is extended to cases where observation patterns may differ across individuals.

  Based on the grouping introduced in Section 2.3.2, each observation pattern belongs to one of the two disjoint sets A and B. For a group $g \in \{A, B\}$, the group–specific mean function $\mu_g(t)$ and observation probability $p_g(t)$ are estimated from the available data by

  $$\hat{p}_g(t) = \frac{1}{n}\sum_{i=1}^{n} O_i(t)\,\mathbb{1}\{O_i \in g\}, \qquad \hat{\mu}_g(t) = \frac{1}{n}\frac{1}{\hat{p}_g(t)}\sum_{i=1}^{n} X_i(t)\,O_i(t)\,\mathbb{1}\{O_i \in g\}.$$

  Normalization by $\hat{p}_g(t)$ adjusts the partial sum to the correct group–specific average over the actually observed units. The indicator $\mathbb{1}\{O_i \in g\}$ ensures that only observations belonging to group g contribute to the sum, while all others are automatically excluded. An illustrative example demonstrating the computation of these estimators is provided in Appendix A.2. Under the MCAR assumption, X and O are independent, implying that the corresponding group means satisfy $\mu_A(t) \approx \mu_B(t)$ for all $t \in [0,1]$. Hence, the estimators $\hat{\mu}_A(t)$ and $\hat{\mu}_B(t)$ should be close, and systematic deviations between them indicate violations of MCAR.

- Step 2: Given that X is $L^2[0,1]$-valued, both X and the partially observed XO can then be treated as random elements in the Hilbert space $L^2[0,1]$; see Hsing & Eubank (2013, Theorem 7.1.2 and 7.4.1) for details. This motivates the test statistic

  $$T_{\mu,L^2} = n\,\|\hat{\mu}_A - \hat{\mu}_B\|_{L^2}^2, \qquad \text{where } \|f\|_{L^2}^2 = \int_0^1 f(t)^2\,dt.$$

  This statistic satisfies the weak convergence, i.e., convergence in distribution (Bosq, 2000, Theorem 2.7); see also Hsing & Eubank (2013, Theorem 7.7.6),

  $$T_{\mu,L^2} \xrightarrow{d} \int_0^1 Z(t)^2\,dt,$$

where Z is a centered Gaussian process with covariance function $k(s,t)$. The integral on the right-hand side can be further expressed as a weighted chi-square distribution,

$$\int_0^1 Z(t)^2\,dt \stackrel{d}{=} \sum_{j=1}^{\infty} \lambda_j \chi^2_{1;j},$$

where $(\chi^2_{1;j})_j$ are independent chi-square distributed random variables with one degree of freedom, and $(\lambda_j)_j$ are the eigenvalues of the covariance operator associated with the kernel $k(s,t)$. For further background on the broader theoretical foundation, we refer to Ofner et al. (2025), Bosq (2000), Hsing & Eubank (2013), and Billingsley (1999).

- Step 3: Building on the general definition introduced in Section 2.1.2, the estimation of the covariance function $k(s,t)$ must take into account that individual trajectories may be only partially observed. A direct plug-in of the general formula is not feasible since the product $(X_i(s) - \mu(s))(X_i(t) - \mu(t))$ is undefined whenever either $X_i(s)$ or $X_i(t)$ is missing. Restricting the computation to available pairs would yield a biased estimator because the probability of observing both s and t may vary over time. Additional theoretical background and a detailed discussion of this correction are provided in Ofner et al. (2025, Section 3.1, p. 7, and Section 5.1, p. 12).

  To correct for this, we weight the observed pairs by their estimated observation probabilities, analogous to the adjustment used for the group–specific means in Step 1. Using the grouping introduced in Section 2.3.2, each observation pattern belongs to one of the two disjoint sets A and B. For $g \in \{A, B\}$ and $t \in [0,1]$, we define the centered functions

$$\tilde{X}_i(t) = X_i(t) - \big(\hat{\mu}_A(t)\,\mathbb{1}\{O_i \in A\} + \hat{\mu}_B(t)\,\mathbb{1}\{O_i \in B\}\big),$$

  and the estimated observation probabilities $\hat{p}_g(t)$ as in Step 1. The resulting estimator of $k(s,t)$ is then given by

$$\hat{k}(s,t) = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{\tilde{X}_i(s)\tilde{X}_i(t)\,O_i(s)O_i(t)\,\mathbb{1}\{O_i \in A\}}{\hat{p}_A(s)\hat{p}_A(t)} + \frac{\tilde{X}_i(s)\tilde{X}_i(t)\,O_i(s)O_i(t)\,\mathbb{1}\{O_i \in B\}}{\hat{p}_B(s)\hat{p}_B(t)}\right).$$

- Step 4: For feasible computing, the infinite weighted chi-square distribution $\sum_{j=1}^{\infty} \lambda_j \chi^2_{1;j}$ is approximated by a finite sum $\sum_{j=1}^{q} \hat{\lambda}_j \chi^2_{1;j}$, where q acts as a truncation parameter. In practice, q is commonly selected adaptively, for example, using the *fraction of variance explained* (FVE) criterion.

  Here, $\hat{\lambda}_j$ denotes the j-th eigenvalue of the covariance estimator $\hat{k}(s,t)$, sorted from the largest eigenvalue to the smallest. The FVE criterion selects q as the smallest integer such that the first q eigenvalues explain a sufficiently large proportion of the total variance, typically 95% or 99%. See Jolliffe (2002, Chapter 6) for a general discussion of variance-based criteria for selecting the number of principal components (via their eigenvalues):

$$\mathrm{FVE}(q) = \frac{\sum_{j=1}^{q} \hat{\lambda}_j}{\sum_{j=1}^{\infty} \hat{\lambda}_j}.$$

- Step 5: For each $b = 1, \dots, B^*$, draw q standard normals and compute $W^{(b)}$ as a weighted sum of their squares, with weights given by the estimated eigenvalues.

- Step 6: Approximate the p-value as

$$p = \frac{\sum_{b=1}^{B^*} \mathbb{1}\{W^{(b)} \geq T_{\mu,L^2}\} + 1}{B^* + 1},$$

where

$$\mathbb{1}\{W^{(b)} \geq T_{\mu,L^2}\} = \begin{cases} 1, & \text{if } W^{(b)} \geq T_{\mu,L^2}, \\ 0, & \text{otherwise.} \end{cases}$$

i.e., the proportion of simulated values $W^{(b)}$ that are greater than or equal to the observed test statistic. In practice, a Monte Carlo correction is applied by slightly adjusting the empirical tail probability. This prevents the estimated p-value from being exactly zero, which would incorrectly suggest that observing such an extreme statistic is *impossible* under H_0. The correction ensures that even if no simulated value exceeds the observed test statistic, the resulting p-value remains strictly positive and retains the correct frequentist error-rate interpretation (Davison & Hinkley, 1997).

Algorithm 1 Asymptotic T_{μ,L^2} test

1: Estimate group-specific means and compute T_{μ,L^2} from $(X_i, O_i)_{i=1}^n$;
2: Estimate the covariance function k by $\hat{k}$;
3: Compute the first q eigenvalues $(\hat{\lambda}_j)_{j=1}^q$ of $\hat{k}$;
4: **for** $b = 1, \dots, B^*$ **do**
5: Sample $Z_1^{(b)}, \dots, Z_q^{(b)} \overset{\text{i.i.d.}}{\sim} N(0, 1)$;
6: Compute $W^{(b)} = \sum_{j=1}^q (Z_j^{(b)})^2 \hat{\lambda}_j$;
7: **end for**
8: Approximate the p-value using T_{μ,L^2} and $W^{(1)}, \dots, W^{(B^*)}$.

2.3.4 Mean Comparison via Supremum Test

Compared to the L^2-based test, the supremum-based test relies on a different norm for comparing the mean functions. While the L^2 test measures the average squared distance between the group-specific mean functions, the supremum test focuses on the largest pointwise deviation. This makes the supremum test more sensitive to local differences that may be averaged out in the L^2 case. Since the observed product XO may contain jumps due to missingness, the supremum test requires working in the Skorokhod space $D[0, 1]$, as the sup-norm is sensitive to such discontinuities. In contrast, the L^2 test is less affected because it averages deviations over the domain, thereby smoothing out local jumps. The supremum approach also provides a natural path for constructing simultaneous confidence bands. On the downside, the method is computationally more demanding, since both eigenvalues and eigenfunctions of the covariance operator must be estimated,

and it requires stronger assumptions on the underlying process. For a more thorough explanation, we refer to Section 3.1 (pp. 8–9) of Ofner et al. (2025).
The null hypothesis remains the equality of mean functions:

$$H_0^{\mu} : \mathbb{E}[X(t) \mid A] = \mathbb{E}[X(t) \mid B], \qquad \forall\, t \in [0,1].$$

Step-by-step outline of the asymptotic-mean supremum test; see Algorithm 2:

- Step 1: Estimate group-specific mean functions $\hat{\mu}_A$ and $\hat{\mu}_B$.
- Step 2: It is of interest to consider alternative function spaces beyond $L^2[0,1]$, as they differ not only in their norms but also in the types of functions they accommodate. In $L^2[0,1]$, the analysis benefits from its Hilbert-space properties and the flexibility to include functions that may be discontinuous or even unbounded, as long as their squared integral remains finite. However, it mainly captures average, mean-squared deviations between groups. In contrast, functions in $C[0,1]$ are continuous and therefore bounded on $[0,1]$, which makes the sup-norm a natural measure for capturing the maximum deviation between functions. It is therefore beneficial when local peaks, short-term differences, or simultaneous confidence bands for the mean difference are of primary interest. Nevertheless, the pointwise product XO may contain jumps even if X itself is continuous. Consequently, the asymptotic analysis is more suitably formulated in the Skorokhod space $D[0,1]$ of càdlàg functions rather than in $C[0,1]$(Billingsley, 1999, p. 135). Working in $D[0,1]$ is more restrictive than in the Hilbert space $L^2[0,1]$, as it requires additional assumptions to establish a framework that includes conditions such as weak convergence (Ofner et al., 2025, p. 8). These considerations motivate the definition of the test statistic

$$T_{\mu,D} = \sqrt{n}\,\|\hat{\mu}_A - \hat{\mu}_B\|_\infty, \qquad \text{where } \|f\|_\infty = \sup_{t\in[0,1]} |f(t)|.$$

- Step 3: Estimate the covariance function using $\hat{k}$.
- Step 4: Compute the first q eigenvalues $(\hat{\lambda}_j)_{j=1}^q$ and eigenfunctions $(\hat{\phi}_j)_{j=1}^q$ of $\hat{k}$.
- Step 5: For each $b = 1, \dots, B^*$, draw q standard normals and compute $W^{(b)}$ as the supremum of the weighted sum over $t \in [0,1]$, using the estimated eigenvalues and eigenfunctions.
- Step 6: Approximate the p-value as

$$p = \frac{\sum_{b=1}^{B^*} \mathbb{1}\{W^{(b)} \geq T_{\mu,D}\} + 1}{B^* + 1},$$

where

$$\mathbb{1}\{W^{(b)} \geq T_{\mu,D}\} = \begin{cases} 1, & \text{if } W^{(b)} \geq T_{\mu,D}, \\ 0, & \text{otherwise.} \end{cases}$$

i.e., the proportion of simulated values $W^{(b)}$ that are greater than or equal to the observed test statistic. As above, the p-value is corrected to remain strictly positive and maintain the correct error-rate interpretation (Davison & Hinkley, 1997).

Algorithm 2 Asymptotic $T_{\mu,D}$ test

1: Estimate group-specific means and compute $T_{\mu,D}$ from $(X_i, O_i)_{i=1}^n$;
2: Estimate the covariance function k by $\hat{k}$;
3: Compute the first q eigenvalues $(\hat{\lambda}_j)_{j=1}^q$ and eigenfunctions $(\hat{\phi}_j)_{j=1}^q$ of $\hat{k}$;
4: **for** $b = 1, \dots, B^*$ **do**
5: Sample $Z_1^{(b)}, \dots, Z_q^{(b)} \overset{\text{i.i.d.}}{\sim} N(0,1)$;
6: Compute $W^{(b)} = \sup_{t\in[0,1]} \left| \sum_{j=1}^q Z_j^{(b)} \sqrt{\hat{\lambda}_j}\, \hat{\phi}_j(t) \right|$;
7: **end for**
8: Approximate the p-value using $T_{\mu,D}$ and $W^{(1)}, \dots, W^{(B^*)}$.

2.3.5 Confidence Bands

A key advantage of the supremum approach in Section 2.3.4 is that one can directly derive simultaneous confidence bands for the mean difference curve $\hat{\mu}_A - \hat{\mu}_B$. Instead of only approximating a p-value, the simulated values of the test statistic can be used to construct $(1-\alpha)$ simultaneous confidence bands over $t \in [0,1]$. The procedure is summarized in Algorithm 3.

Algorithm 3 Simultaneous confidence bands for $\hat{\mu}_A - \hat{\mu}_B$

1: Compute $\hat{\mu}_A$ and $\hat{\mu}_B$ from $(X_i, O_i)_{i=1}^n$;
2: Estimate the covariance function k by $\hat{k}$;
3: Compute the first q eigenvalues $(\hat{\lambda}_j)_{j=1}^q$ and eigenfunctions $(\hat{\phi}_j)_{j=1}^q$ of $\hat{k}$;
4: **for** $b = 1, \dots, B^*$ **do**
5: Sample $Z_1^{(b)}, \dots, Z_q^{(b)} \overset{\text{i.i.d.}}{\sim} N(0,1)$;
6: Compute $W^{(b)} = \sup_{t\in[0,1]} \left| \sum_{j=1}^q Z_j^{(b)} \sqrt{\hat{\lambda}_j}\, \hat{\phi}_j(t) \right|$;
7: **end for**
8: Compute the empirical $(1-\alpha)$-quantile $\hat{q}_\alpha$ of $W^{(1)}, \dots, W^{(B^*)}$;
9: Construct the simultaneous confidence band $\hat{\mu}_A(t) - \hat{\mu}_B(t) \pm \hat{q}_\alpha/\sqrt{n}$.

2.3.6 Mean Comparison via Bootstrap

While the asymptotic theory provides a principled way to approximate the distribution of the test statistics, it can be numerically challenging and relies on additional assumptions (e.g. specific conditions in the Skorokhod space). Bootstrap procedures offer a practical and widely applicable alternative. They approximate the distribution of a test statistic by resampling the data. This has several advantages: flexible adaptation to finite-sample structure, no need for eigen-decomposition, and improved small-sample performance. For a more detailed background of bootstrap approximations, we refer to Ofner et al. (2025, Section B, pp. 22–23).
In our setting, resampling is done *groupwise*, i.e. separately within each group A and B, with respect to the null hypothesis of equal means. Within this framework, we derive two bootstrap procedures: a mean test and simultaneous confidence bands. These serve as an

alternative to the asymptotic T_{μ,L^2} and $T_{\mu,D}$ tests as well as their associated confidence bands.
The null hypothesis remains the same:

$$H_0^{\mu} : \mathbb{E}[X(t) \mid A] = \mathbb{E}[X(t) \mid B], \qquad \forall\, t \in [0,1].$$

Step-by-step outline of the bootstrap mean comparison test; see Algorithm 4:

- Step 1: Estimate group-specific mean functions $\hat{\mu}_A$ and $\hat{\mu}_B$.
- Step 2: Compute the observed test statistic T (T_{μ,L^2} or $T_{\mu,D}$) from $(X_i, O_i)_{i=1}^n$.
- Step 3: Compute the group sizes n_A, n_B and construct centered datasets $(X_i - \hat{\mu}_A, O_i)$ for A and $(X_i - \hat{\mu}_B, O_i)$ for B.
- Step 4: For each bootstrap sample $b = 1, \ldots, B^*$, resample n_A pairs from group A and n_B pairs from group B with replacement, and compute the bootstrap statistic $T^{(b)}$.
- Step 5: Approximate the p-value as described in T_{μ,L^2} or $T_{\mu,D}$.

Algorithm 4 Bootstrap approximation for generic mean comparison T (either T_{μ,L^2} or $T_{\mu,D}$)

```
Compute μ̂_A, μ̂_B and the test statistic T from (X_i, O_i)_{i=1}^n;
Compute n_A = Σ_{i=1}^n 1{O_i ∈ A} and n_B = Σ_{i=1}^n 1{O_i ∈ B};
Create a data set (X_j^A, O_j^A)_{j=1}^{n_A} containing the (X_i − μ̂_A, O_i) with O_i ∈ A;
Create a data set(X_j^B, O_j^B)_{j=1}^{n_B} containing the (X_i − μ̂_B, O_i) with O_i ∈ B;
for b = 1, ..., B* do
    Sample (X_j^{A(b)}, O_j^{A(b)})_{j=1}^{n_A} from (X_j^A, O_j^A)_{j=1}^{n_A} with replacement;
    Sample (X_j^{B(b)}, O_j^{B(b)})_{j=1}^{n_B} from (X_j^B, O_j^B)_{j=1}^{n_B} with replacement;
    Compute T^(b) from (X_j^{A(b)}, O_j^{A(b)})_{j=1}^{n_A} and (X_j^{B(b)}, O_j^{B(b)})_{j=1}^{n_B};
end for
Approximate the p-value using T and T^(1), ..., T^(B*).
```

2.3.7 Confidence Bands via Bootstrap

Beyond hypothesis testing, the bootstrap approach can also be employed to construct simultaneous confidence bands for the mean difference $\hat{\mu}_A - \hat{\mu}_B$. The idea parallels the supremum-based approach, but the critical quantile is obtained from bootstrap resamples instead of the weighted supremum of standard normals. For further insight, we refer to Algorithm 5.

Algorithm 5 Bootstrap confidence bands for $\hat{\mu}_A - \hat{\mu}_B$

Compute $n_A = \sum_{i=1}^{n} \mathbb{1}\{O_i \in A\}$ and $n_B = \sum_{i=1}^{n} \mathbb{1}\{O_i \in B\}$;
Create a data set $(X_j^A, O_j^A)_{j=1}^{n_A}$ containing the $(X_i - \hat{\mu}_A, O_i)$ with $O_i \in A$;
Create a data set $(X_j^B, O_j^B)_{j=1}^{n_B}$ containing the $(X_i - \hat{\mu}_B, O_i)$ with $O_i \in B$;
for $b = 1, \dots, B^*$ **do**
    Sample $(X_j^{A(b)}, O_j^{A(b)})_{j=1}^{n_A}$ from $(X_j^A, O_j^A)_{j=1}^{n_A}$ with replacement;
    Sample $(X_j^{B(b)}, O_j^{B(b)})_{j=1}^{n_B}$ from $(X_j^B, O_j^B)_{j=1}^{n_B}$ with replacement;
    Compute $T_{\mu,D}^{(b)}$ from $(X_j^{A(b)}, O_j^{A(b)})_{j=1}^{n_A}$ and $(X_j^{B(b)}, O_j^{B(b)})_{j=1}^{n_B}$;
end for
Compute the empirical $(1-\alpha)$-quantile $\hat{q}_\alpha$ of $T_{\mu,D}^{(1)}, \dots, T_{\mu,D}^{(B^*)}$;
Construct the simultaneous confidence band $\hat{\mu}_A - \hat{\mu}_B \pm \hat{q}_\alpha / \sqrt{n}$.

2.3.8 Implementation in R

The implementation of the proposed tests is written in `R` and is openly available via a GitHub repository, see Section B. It relies on the package `tf` (Scheipl & Goldsmith, 2025), which provides the core functionality for representing and handling functional data, and on `tidyfun` (Scheipl et al., 2025), which extends this framework by enabling data manipulation using `tidyverse` tools. Efficient parallelization and reproducible resampling are achieved with the packages `foreach` (Microsoft & Weston, 2022), `doParallel` (Corporation & Weston, 2022), and `doRNG` (Gaujoux, 2025). For fast and reliable argument checking, the package `checkmate` (Lang, 2017) is employed. In addition, functions from the base R distribution, including `stats` and `parallel` (R Core Team, 2023), are used for statistical computations, random number generation, and general utilities. Together, this software stack provides a reproducible environment for implementing and evaluating the proposed MCAR tests.

Common arguments shared by all implemented algorithms

Argument	**Default**	**Explanation**
`fd`	`NULL`	Functional data in `tfd` format (from `tidyfun`). If supplied, takes precedence over `X`.
`X`	`NULL`	Functional data provided as a dense $n \times m$ numeric matrix with `NA`s indicating unobserved entries. This serves as an alternative input format when no `fd` object is used.
`groups`	`NULL`	Two-level grouping vector of length n, assigning each curve either to group A (complete/more observed) or group B (incomplete/less observed). If omitted, grouping is created automatically via `observed_ratio`. Note: when groups are specified manually, labels are swapped (if necessary) so that group A always corresponds to the more complete sample.
`observed_ratio`	`1`	Coverage threshold $\delta \in [0, 1]$ used for auto-grouping when `groups` is not provided (see Section 2.3.2). Curves with observation ratio $\geq \delta$ are assigned to A, the rest to B.
`min_frac`	`0.10`	Restricts analysis to time points t_j where both groups have at least $n \times$ `min_frac` observed values (following the subdomain definition in Section 2.3.2).
`n_sim/n_boot`	`10000`	Number of Monte Carlo draws (L^2/supremum) or bootstrap replications.
`seed`	`NULL`	RNG seed; in parallel mode, reproducible L'Ecuyer–CMRG streams are created across workers via `doRNG`.

Table 3: Common arguments used across all functions.

1) Asymptotic L^2 test

```
asym_mean_L2_test(fd, X, groups, observed_ratio, min_frac, n_sim,
                  seed, fve)
```

Argument	Default	Explanation
fve	0.99	Used to select the smallest number q of eigenvalues which explain at least 99% of the total variance. FVE is described in Step 4 of Section 2.3.3.

Table 4: Additional argument for the asymptotic L^2 test.

2) Asymptotic supremum test

```
asym_mean_sup_test(fd, X, groups, observed_ratio, min_frac, n_sim,
                   seed, fve, alpha, compute_bands, bands_only)
```

Argument	Default	Explanation
fve	0.99	Used to select the smallest number q of eigenvalues together with their corresponding eigenfunctions which explain at least 99% of the total variance. FVE is described in Step 4 of Section 2.3.3.
alpha	0.05	Significance level for the simultaneous confidence bands.
compute_bands	TRUE	Return simultaneous confidence bands.
bands_only	FALSE	Only returns the simultaneous confidence bands instead of a full `htest`. Corresponds to Algorithm 3 in Ofner et al. (2025).

Table 5: Additional arguments for the asymptotic supremum test.

3) Bootstrap approximation (L^2 or supremum)

```
boot_mean_test(fd, X, groups, observed_ratio, min_frac, n_boot,
               seed, alpha, compute_bands, bands_only, ncpus, stat,
               chunk_size, manage_backend, worker_blas_threads,
               max_redraws)
```

Argument	Default	Explanation
`alpha`	`0.05`	Significance level for simultaneous confidence bands (relevant only if `stat = "D"` and `compute_bands = TRUE`).
`compute_bands`	`TRUE`	Compute confidence bands. Only applicable for the supremum statistic "`D`"; ignored for `L2`.
`bands_only`	`FALSE`	Only returns the simultaneous confidence bands instead of a full `htest`. Corresponds to Algorithm 7 in Ofner et al. (2025).
`ncpus`	`detectCores(TRUE) - 2`	Number of parallel workers; by default set to the number of available logical cores (`parallel::detectCores(TRUE)`) minus two, but not less than one.
`stat`	`c("L2", "D")`	Statistic: `"L2"`, `"D"` (supremum), or both.
`chunk_size`	`NULL`	Number of bootstrap replicates grouped into one parallel task. Larger chunks reduce scheduling overhead but risk load imbalance; smaller chunks balance better across workers but increase overhead. If `NULL`, a heuristic selects a default that scales with the total number of bootstrap replicates (`n_boot`) and available workers (`ncpus`), typically resulting in chunk sizes between 50 and 200 replicates depending on the workload.
`manage_backend`	`"auto"`	How to handle the parallel backend: `"auto"` reuses an existing `foreach` backend if available, otherwise creates an internal pool; `"force_pool"` always creates a fresh internal pool; `"sequential"` disables parallelization.
`worker_blas_threads`	`1`	Number of BLAS/OpenMP threads *per worker*. Advanced users may increase this value, but only if: $$n_{\text{cpus}} \times \text{worker\_blas\_threads} \leq \text{ available CPU cores}$$ The number of available CPU cores can be obtained via `parallel::detectCores(TRUE)`. Otherwise, if all CPU cores are already occupied, additional BLAS threads have no free cores and only add scheduling overhead. In practice, values greater than `1` are only beneficial for a few costly matrix operations when no parallel workers are used.
`max_redraws`	`20`	Maximum number of redraw attempts for invalid bootstrap samples. A bootstrap sample is considered invalid if one of the groups has zero observations at any time point. If this limit is reached, the replicate is skipped with a warning. Increasing this value can improve robustness for very sparse or highly censored data but increases runtime.

Table 6: Additional arguments for the bootstrap mean test.

3 Empirical Evaluation

To ensure that the proposed tests perform as intended and yield reproducible results, we subject them to a series of evaluations, starting with a simulation study and followed by applications to real data. All simulations are implemented in `R`, with details given in Section 2.3.8. Unless stated otherwise, we assume that curves are measured on an equispaced grid and use 10,000 samples to approximate the distribution of the test statistics under H_0 in both the asymptotic and the bootstrap settings.

3.1 Replication: Simulation Study

First, a controlled simulation study is conducted to verify that the proposed tests operate as intended. We investigate both missingness mechanisms, starting with MCAR. For this purpose, we simulate $X_1, \dots, X_n$ as independent realizations of a standard Brownian motion on $[0, 1]$, observed on an equispaced grid of $p = 100$ points. To assess whether the tests achieve the nominal significance level $\alpha = 0.05$, we estimate the type I error based on 5,000 simulation runs with sample sizes $n \in \{100, 250, 500\}$.

Case 1: Missing completely at random (MCAR)

Under MCAR the observation process O is independent of X and generated as:

$$O(t) = \begin{cases} 1, & \text{with prob. } 0.5, \\ \mathbb{1}_{[L,U)}(t), & \text{with prob. } 0.5, \end{cases} \qquad t \in [0,1],$$

where $L = \min\{U_1, U_2\}$, $U = \max\{U_1, U_2\}$ and $U_1, U_2 \overset{\text{i.i.d.}}{\sim} \text{Unif}[0,1]$. Curves are partitioned after Partition 1, yielding a complete and an incomplete group. The type I error probabilities reported by Ofner et al. (2025, Sec. 5.2) are shown in Table 7, while the corresponding results from our implementation, are presented in Table 8. Across n and for both the asymptotic and the bootstrap approximations, the results closely resemble the published values and approach the nominal level $\alpha = 0.05$, in some cases matching it exactly.

Case 2: Missing not at random (MNAR/censoring)

Next, we examine departures from MCAR by imposing censoring according to,

$$O(t) = \mathbb{1}\{X(t) \in (a,b)\}, \qquad t \in [0,1],$$

with $a = -1$ and $b \in \{1.00, 1.10, \ldots, 2.00\}$. The curves are grouped again according to Partition 1. To examine the influence of censoring, we begin with the symmetric case obtained by setting $b = 1$, which yields $X(t) \in (-1, 1)$. Since the mean functions coincide, i.e. $\mu_A(t) = \mu_B(t) = 0$, T_{μ,L^2} and $T_{\mu,D}$ exhibit low discriminatory power, as the tests are designed to detect mean differences. Increasing the value of b breaks the symmetry of the censoring mechanism, leading to an MNAR pattern that can be reliably detected by our tests. As in Case 1, rejection probabilities are estimated based on 5,000 simulation runs for each configuration of (a, b). As shown in Figure 3, both tests remain close to the significance level $\alpha = 0.05$ at $(a, b) = (-1, 1)$ and require some degree of asymmetry to start identifying the underlying MNAR pattern. They exhibit steadily increasing rejection probabilities with larger b. In Figure 4, a violation of H_0 is visible on the interval $t \in [0.40, 1]$, indicated by the 95% simultaneous confidence band for $\mu_A(t) - \mu_B(t)$, which no longer contains zero. In this region, the mean of the complete group is significantly higher than that of the incomplete group. The confidence bandd is obtained via Algorithm 3.

	asymptotic		bootstrap	
n	T_{μ,L^2}	$T_{\mu,D}$	T_{μ,L^2}	$T_{\mu,D}$
100	0.07	0.07	0.06	0.06
250	0.06	0.07	0.06	0.06
500	0.05	0.05	0.05	0.05

Table 7: Estimated type I error probabilities in Case 1 (MCAR) for both tests at $\alpha = 0.05$.

	asymptotic		bootstrap	
n	T_{μ,L^2}	$T_{\mu,D}$	T_{μ,L^2}	$T_{\mu,D}$
100	0.06	0.06	0.05	0.04
250	0.05	0.06	0.05	0.05
500	0.06	0.06	0.06	0.05

Table 8: Replication results for estimated type I error probabilities in Case 1 (MCAR) for both tests at $\alpha = 0.05$.

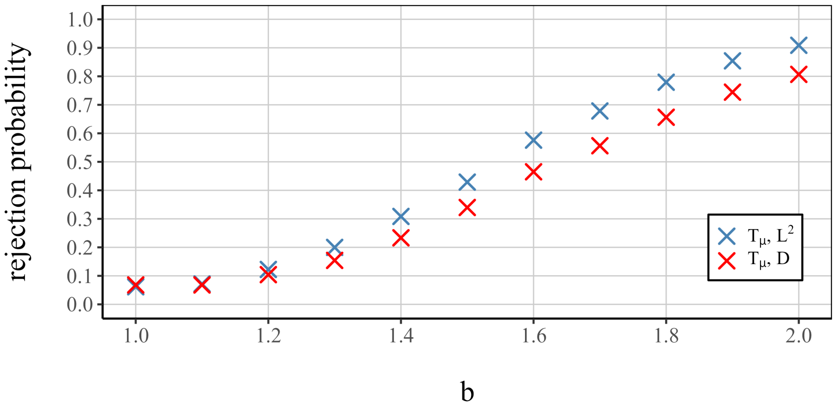

Figure 3: Case 2 (MNAR): rejection probabilities of the asymptotic tests T_{μ,L^2} and $T_{\mu,D}$ for $n = 100$. Curves are observed only if $X(t) \in (a, b)$ with $a = -1$ and varying b. Larger values of b increase the asymmetry of the censoring mechanism.

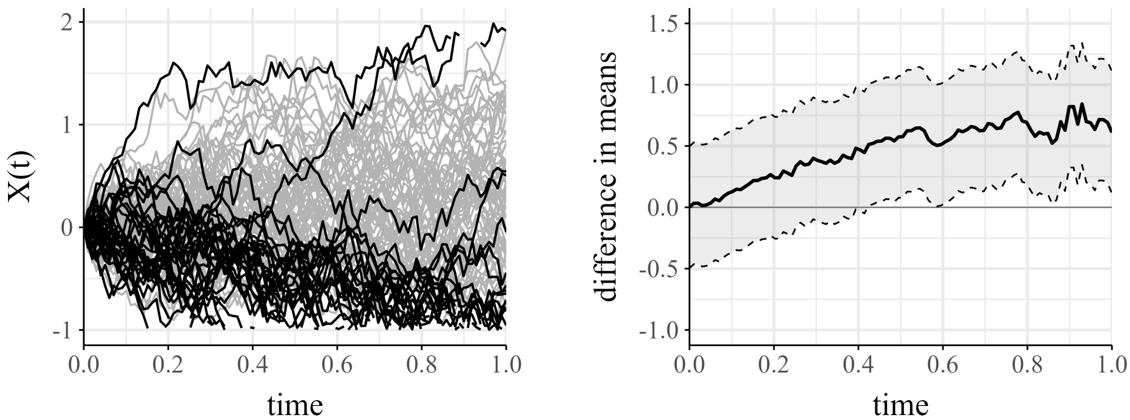

Figure 4: Case 2 (MNAR) with asymmetric censoring: $n = 100$ Brownian motion paths (left) and the estimated mean difference $\hat{\mu}_A - \hat{\mu}_B$ with 95% simultaneous confidence band (right). Curves are observed only if $X(t) \in (-1, 2)$.

3.2 Replication: Real Data Applications

3.2.1 Heart rate profiles

As a first real-data example, we revisit partially observed heart rate curves from the "Swiss Kidney Project on Genes in Hypertension" (Alwan et al., 2014), previously analyzed in Kraus (2015, 2019). The data consists of $n = 878$ heart rate profiles measured on an equispaced grid of $p = 361$ time points within the interval [8 pm, 2 am]. For subject i, $X_i(t)$ denotes the heart rate (in beats per minute) at time t. Among the sample, 527 curves (around 60%) are fully observed.
As reported by Kraus (2015), missingness in this dataset is mainly caused by participants removing the monitoring devices due to discomfort or by technical device failures. Since these reasons are unrelated to the underlying heart rate process itself, the MCAR assumption appears plausible in this context. To evaluate this formally, we partition the data into complete and incomplete functions as in Partition 1. Applying our tests does not reveal significant differences between the two groups, which further supports the MCAR hypothesis. The obtained p-values are $p_{\mu,L^2} = 0.63$ and $p_{\mu,D} = 0.64$. For additional details on this case study, we refer to Section 5.3 (p. 14) of Ofner et al. (2025). Figure 5 illustrates the heart rate profiles and displays the estimated difference $\hat{\mu}_A - \hat{\mu}_B$ together with the simultaneous confidence band obtained from Algorithm 3. The mean difference function remains close to zero and shows no notable deviations. This plot reproduces the original figure presented in Ofner et al. (2025, Figure 6, p. 17).

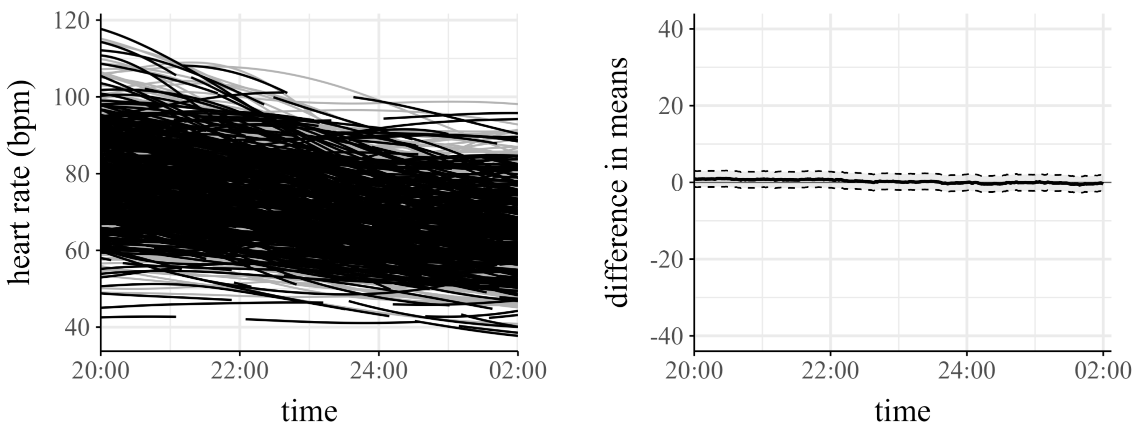

Figure 5: Observed heart rate profiles (left) and estimated mean difference $\hat{\mu}_A - \hat{\mu}_B$ with 95% simultaneous confidence band (right).

3.2.2 Electricity data

The basis for our second analysis is a dataset of the german electricity market, previously introduced by Liebl & Rameseder (2019). The dataset consists of $n = 608$ price curves X_i, each describing the logarithmized electricity price (EURO/MW) as a function of demand (MW). We restrict attention to the interval $[1700, 2500]$ MW, represented by 161 equispaced grid points, where incomplete observations occur. Each curve is available over the initial part of the domain but may drop out in the later part. Once missingness sets in, the corresponding function X_i remains unobserved until the end of the interval. Only a few of the electricity curves cover the entire domain, which makes the partition into complete and incomplete functions (as in Partition 1) impractical, since it would result in highly unbalanced groups. Instead, we classify the data according to the size of the observed intervals and apply Partition 2, with $\delta = 0.43$, which divides the sample into two groups of approximately equal size. To meet the requirements of our Subdomain definition, meaning that at each grid point a sufficient proportion of curves from both groups are observed, we restrict our analysis to the interval $I = [1700, 2000]$ MW. Due to the properties of the electricity market, curves spanning a larger part of the domain appear to be associated with higher price levels, suggesting a potential violation of the MCAR assumption. Applying our tests T_{μ,L^2} and $T_{\mu,D}$ confirms this concern, as the MCAR hypothesis is clearly rejected. An illustration of the data and its mean difference curve is given in Figure 6, together with the corresponding 95% simultaneous confidence band. The band is calculated as depicted in Algorithm 3. The band shows a clear deviation from the zero line across the entire selected subdomain $I = [1700, 2000]$ MW. Further details on this application, as well as the corresponding plot for comparison, are provided in Section 5.4 (pp. 16–17) of Ofner et al. (2025).

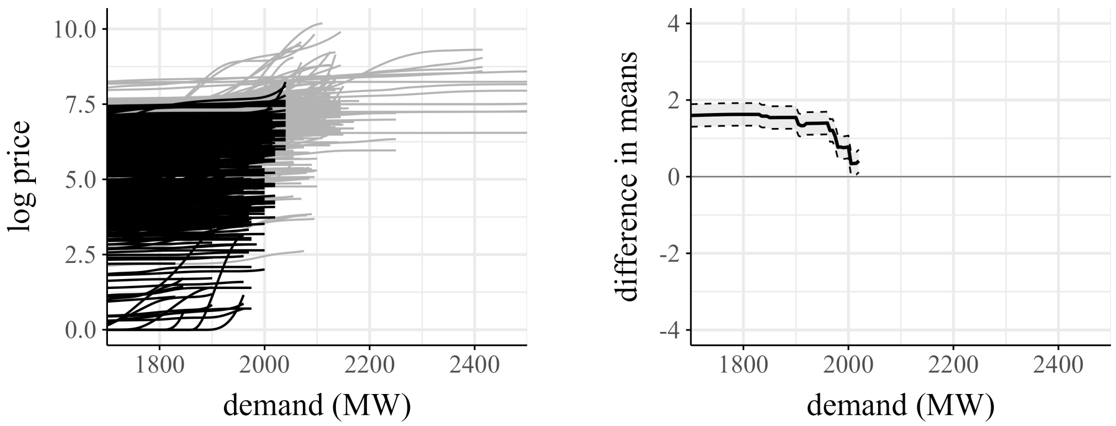

Figure 6: Observed electricity curves (left) and estimated mean difference $\hat{\mu}_A - \hat{\mu}_B$ with 95% simultaneous confidence band (right).

3.2.3 Temperature data

Third, we analyze temperature curves recorded every half hour in Graz (Austria), provided by Land Steiermark (2023). The dataset consists of $n = 76$ daily curves, of which 66 are fully observed, while the remaining 10 show missing values in the second half of the day, specifically for $t \in [15{:}00, 17{:}00]$. An illustration of the data is given in Figure 7. For comparison, this corresponds to Figure 6 (p. 17) in Ofner et al. (2025).
According to the monitoring agency, the cause of missingness is not entirely clear. It could either be due to the sensors' sensitivity to high temperatures or a failure in the transmission unit. In the case of temperature sensitivity, this would correspond to censoring and thus imply a violation of MCAR. For the partition, we again define two groups

A and B, following Partition 1. The relatively small sample size poses additional challenges, in particular because the incomplete group B does not cover the full domain (see $t \in [15{:}00, 17{:}00]$). To ensure valid inference, it is therefore necessary to restrict the analysis to a suitable subdomain $I \subseteq [0,1]$, as described in Section 2.3.2. To evaluate the influence of this choice, different subgrids are examined by selecting $m \in \{24, \ldots, 45\}$ grid points with the highest number of non-missing observations. Due to the limited sample size, the test distributions are approximated using bootstrap procedures with $B^* = 10{,}000$ repetitions instead of asymptotic approximations. The same analysis can be found in Ofner et al. (2025, Section 5.5, p. 16).

The impact of different subgrid choices on the test results is reported in Figure 8. This plot represents a replication of the original results reported in Ofner et al. (2025, Figure 7, p. 18). The p-values of both bootstrap tests remain stable across the different subgrid choices. Since all p-values are significant, the results provide strong evidence that the temperature sensors are affected by a systematic malfunction, thus violating the MCAR hypothesis.

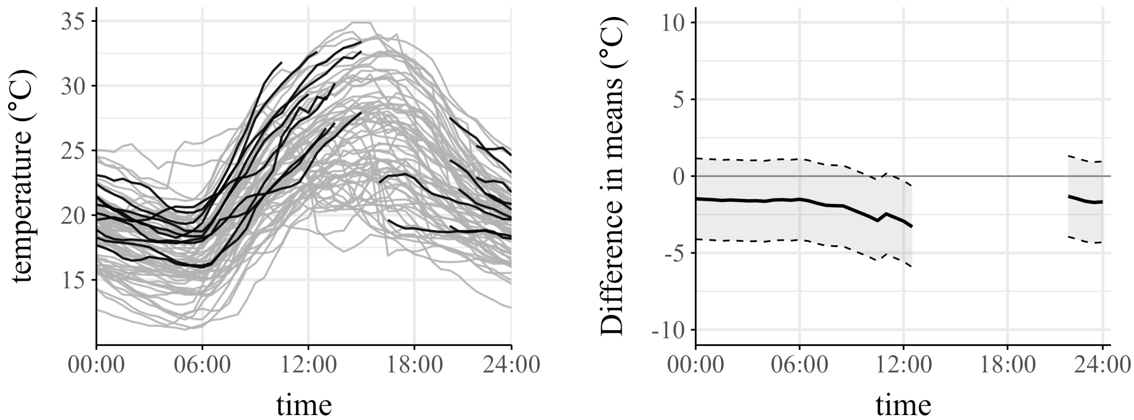

Figure 7: Observed temperature curves (left) and estimated mean difference $\hat{\mu}_A - \hat{\mu}_B$ with 95% simultaneous confidence band (right).

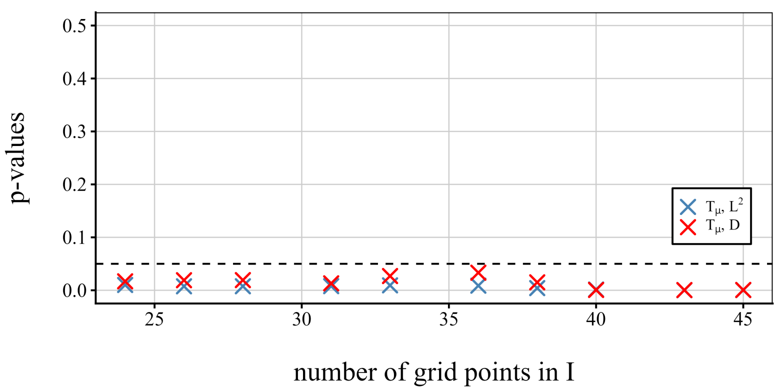

Figure 8: p-values across $m \in \{24, \ldots, 45\}$ grid points within the subdomain $I \subseteq [0,1]$.

4 Discussion

The proposed testing framework examines the MCAR assumption by dividing functional data into two groups and comparing their mean functions. This strategy provides clear hypotheses and practical procedures, but it is also subject to several limitations:

1. **Two-group partition**
   Throughout, we assume a disjoint split and test equality of means across these two groups. While this restriction allows the use of simple testing strategies for missingness as described in Section 2.3, it may limit the ability to obtain more nuanced insights that could emerge from finer partitioning. Extending the methodology to more than two groups could therefore be beneficial. For example, in settings where curves differ not only by being complete or incomplete but also by the extent or timing of missing segments, a multi-group comparison may provide a clearer explanation of systematic differences. This concern has already been highlighted in Ofner et al. (2025) as a natural direction for possible extensions.

2. **Lack of data-adaptivity**
   In our framework, groups are defined by simple, pre-specified rules based on the observation process O (e.g., complete vs. incomplete, or according to the size of the observed domain). While this ensures that the test remains feasible, it does not allow for more sophisticated, data-driven strategies that could automatically detect and separate heterogeneous missingness mechanisms (e.g., subsets that appear closer to MCAR versus those suggestive of MNAR). To the best of our knowledge, no such fully data-adaptive partitioning approaches have been developed for MCAR testing in functional data. Therefore, data-adaptive partitioning strategies could represent a promising direction for future research in MCAR testing, a point also emphasized by Ofner et al. (2025).

3. **Assumption of i.i.d. curves**
   All theoretical results are derived under the assumption that the functional curves X_i are independent and identically distributed. This excludes many practically relevant settings, such as functional time series with serial dependence, panel data, and repeated-measures data with subject effects. The challenges arising from such dependent data have been emphasized in the literature on functional time series. In particular, Hörmann & Kokoszka (2010) discuss weak dependence for functional time series and demonstrate how temporal dependence affects key statistical procedures, such as estimation and inference in FDA. Extending MCAR tests to such dependent settings remains beyond the scope of the current framework, but investigating these extensions would be a worthwhile direction for further work.

4. **Sparse or irregular functional data**
   Our theoretical framework presumes functions observed on sufficiently dense and regular grids, which enables estimation of group means and distributions as outlined in Section 2.3. In many practical applications, however, functional data are sparse or irregularly sampled. In such cases, the direct implementation of our tests becomes challenging, as additional steps are required to obtain reliable estimators of the underlying functions. Recent work by Chen (2023) discusses these issues in the context

of sparsely and irregularly observed functional data and provides methodological tools for estimation under such conditions. Moreover, established reconstruction techniques such as PACE (Yao et al., 2005) offer a practical way to estimate mean and covariance functions from sparse observations. While these existing FDA methods could in principle be combined with MCAR tests, a connection between them remains largely unexplored. Developing extensions that adapt the proposed methodology to sparse or irregular sampling, therefore, represents an important step toward broader applicability.

5. Consistency of the tests

We face a main issue regarding the consistency of our proposed tests. A fully consistent test against the most general alternative H_A : "X and O are not independent" is not attainable, as the following example shows. Given these two processes,

Process A

Let $\delta_1, \delta_2 \overset{\text{i.i.d.}}{\sim}$ Bernoulli(1/2).
For $t \in [0,1]$,

$$X(t) = \delta_1 \mathbb{1}_{[0,1/2]}(t) + \delta_2 \mathbb{1}_{(1/2,1]}(t),$$

$$O(t) = \begin{cases} \mathbb{1}_{[0,1/2]}(t), & \text{if } \delta_2 = 1, \\ \mathbb{1}_{[0,1]}(t), & \text{if } \delta_2 = 0. \end{cases}$$

Process B

Let $\delta_1' \sim$ Bernoulli(1/2).
For $t \in [0,1]$,

$$X'(t) = \delta_1' \mathbb{1}_{[0,1/2]}(t),$$

$$O'(t) = \begin{cases} \mathbb{1}_{[0,1/2]}(t), & \text{with prob. } 0.5, \\ \mathbb{1}_{[0,1]}(t), & \text{with prob. } 0.5. \end{cases}$$

it is impossible to distinguish the distributions generated by (XO, O) and $(X'O', O')$. Nevertheless, the underlying processes differ fundamentally: while (X, O) reflects a mechanism that violates MCAR, (X', O') represents the MCAR setting. This example illustrates a theoretical limitation: full consistency cannot be achieved against the most general alternative $H_A : X \not\perp\!\!\!\perp O$ because certain dependence structures may occur only in unobserved regions. In such cases, the dependence between X and O is not identifiable from the observable pairs (XO, O).

Importantly, the mean-based procedures examine the null hypothesis H_0^μ (as described in Section 2.3.3), which solely test the equality of the mean functions rather than full independence. Consequently, a non-rejection must be interpreted narrowly as the absence of detectable mean differences in the observable domain rather than as evidence for $X \perp\!\!\!\perp O$. However, when group-specific mean differences are present and observed, the statistics T_{μ,L^2} and $T_{\mu,D}$ increase with sample size, yielding consistency against H_0^μ. A more detailed discussion of the consistency of the tests is provided in Ofner et al. (2025, Section 4, pp. 11–12). It is of interest for future research to explore additional approaches to ensure consistency, potentially under broader conditions.

5 Conclusion

This work investigated statistical tests for assessing the Missing Completely at Random (MCAR) assumption in functional data, and replicated as well as validated the methodology proposed by Ofner et al. (2025). To evaluate their performance, we conducted a series of simulations and real-data applications. Overall, the simulation study demonstrated that the empirical performance aligns well with the theoretical properties derived in Ofner et al. (2025). In controlled settings, the type I error closely matched the nominal level across different sample sizes and calibrations. The tests behaved as expected under symmetric censoring with low rejection rates, before steadily gaining discriminatory power once asymmetry was introduced. Moreover, the simulations highlighted the robustness of both the asymptotic and the bootstrap versions of the tests, supporting the correctness of our implementation in `R`. For the empirical applications, the tests confirmed the plausibility of MCAR for heart rate profiles, clearly rejected it in the case of electricity market prices, and pointed to systematic malfunctions in temperature sensor data. These empirical examples underline that the tests are not only of theoretical interest but also provide meaningful insights in applied settings, ranging from biomedical studies to energy economics and environmental monitoring. Furthermore, they emphasize the importance of conducting such tests, since many functional data analyses rely critically on the MCAR assumption as a prerequisite for valid inference. To conclude, the results demonstrate that the proposed procedures offer a valuable framework for formally testing the MCAR assumption in functional data, as well as highlighting the need for further research on handling and testing missingness in functional data analysis.

A Appendix

A.1 Use of AI Tools

1. ChatGPT

- Code implementation and technical support: Assistance with translating algorithms from their mathematical formulation into R code, as well as solving practical issues in the implementation (e.g., setting up simulations, parallelization, etc.).
- Text improvement and formulation assistance: Clarifying theoretical arguments, rephrasing sections for better readability, improving text flow and structure, stylistic refinement.
- Structure of the bachelor's thesis: Support with organizing chapters, outlining sections, and improving the overall coherence of the work.
- Literature research: Support with identifying related work in functional data analysis and missing data literature.

2. Grammarly

- Spell checking, grammar and syntax: Correction of typos, grammar, and punctuation.

3. Quality Assurance

All content generated or improved by AI tools was critically reviewed and independently validated. In most topics above, AI tools offered a starting point that was then expanded upon. In the case of the codebase provided in the linked GitHub repository, the underlying structure was developed with the assistance of AI; however, not a single line of code was used without further inspection and supervision. The same applies to the few, singled-out text passages, which were partially generated with the assistance of ChatGPT. Final responsibility for all content lies with the author.

A.2 Illustrative Example: Estimating the Mean at a Single Time Point

To illustrate the mechanics at a single time point t, consider $n = 10$ curves with five belonging to group A and five to group B. In A, three curves are observed at t with values 2.0, 1.5, and 3.0. In B, three curves are observed at t with values 1.2, 2.4, and 1.8.

Curve i	Group	$O_i(t)$	$X_i(t)$	$X_i(t)\,O_i(t)\,\mathbb{1}\{O_i \in g\}$
1	A	1	2.0	2.0
2	A	0	–	0
3	A	1	1.5	1.5
4	A	1	3.0	3.0
5	A	0	–	0
6	B	1	1.2	1.2
7	B	1	2.4	2.4
8	B	0	–	0
9	B	1	1.8	1.8
10	B	0	–	0

Table 9: Observed and missing values at time t for $n = 10$ curves across groups A and B

For each group $g \in \{A, B\}$, the proportion of observed values $\hat{p}_g(t)$ and the corresponding mean $\hat{\mu}_g(t)$ are calculated as follows:

$$\hat{p}_g(t) = \frac{1}{n}\sum_{i=1}^{n} O_i(t)\,\mathbb{1}\{O_i \in g\}, \qquad \hat{\mu}_g(t) = \frac{1}{n}\frac{1}{\hat{p}_g(t)}\sum_{i=1}^{n} X_i(t)\,O_i(t)\,\mathbb{1}\{O_i \in g\}.$$

Applying these estimators to the example data for groups A and B yields:

For group A,

$$\hat{p}_A(t) = \tfrac{1}{10}(1 + 0 + 1 + 1 + 0) = 0.3,$$

$$\sum_{i=1}^{n} X_i(t)O_i(t)\,\mathbb{1}\{O_i \in A\} = 2.0 + 0 + 1.5 + 3.0 + 0 = 6.5.$$

For group B,

$$\hat{p}_B(t) = \tfrac{1}{10}(1 + 1 + 0 + 1 + 0) = 0.3,$$

$$\sum_{i=1}^{n} X_i(t)O_i(t)\,\mathbb{1}\{O_i \in B\} = 1.2 + 2.4 + 1.8 + 0 + 0 = 5.4.$$

Hence,

$$\begin{aligned} \hat{\mu}_A(t) &= \frac{1}{10}\frac{1}{0.3}(6.5) = 2.17 \\ \hat{\mu}_B(t) &= \frac{1}{10}\frac{1}{0.3}(5.4) = 1.80 \end{aligned} \qquad \Longrightarrow \quad \hat{\mu}_A(t) - \hat{\mu}_B(t) = 2.17 - 1.80 = 0.37$$

By weighting with the proportion $\hat{p}_g(t)$, missing observations (encoded as zeros) do not influence the estimated mean and are simply treated as unobserved.

A.3 Illustrative Example: Estimating the Covariance at Two Time Points

To illustrate the estimation of the covariance function $k(s,t)$, consider again the $n = 10$ curves from Appendix A.2. For simplicity, suppose that both time points s and t are observed for these curves, which are grouped as follows:

	A			B		
Curve i	$X_i(s)$	$X_i(t)$	$O_i(s)O_i(t)$	$X_i(s)$	$X_i(t)$	$O_i(s)O_i(t)$
1	1.8	2.0	1	1.0	1.2	1
3	1.2	1.5	1	2.2	2.4	1
4	2.8	3.0	1	1.5	1.8	1

The group–specific means at s and t are

$$\hat{\mu}_A(s) = 1.93, \quad \hat{\mu}_A(t) = 2.17, \qquad \hat{\mu}_B(s) = 1.57, \quad \hat{\mu}_B(t) = 1.80.$$

We define the centered values $\tilde{X}_i(u)$ for $u \in \{s,t\}$ as

$$\tilde{X}_i(u) = X_i(u) - \big(\hat{\mu}_A(u)\,\mathbb{1}\{O_i \in A\} + \hat{\mu}_B(u)\,\mathbb{1}\{O_i \in B\}\big).$$

The covariance estimator from Step 3 is given by

$$\hat{k}(s,t) = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{\tilde{X}_i(s)\tilde{X}_i(t)\,O_i(s)O_i(t)\,\mathbb{1}\{O_i \in A\}}{\hat{p}_A(s)\hat{p}_A(t)} + \frac{\tilde{X}_i(s)\tilde{X}_i(t)\,O_i(s)O_i(t)\,\mathbb{1}\{O_i \in B\}}{\hat{p}_B(s)\hat{p}_B(t)}\right).$$

Applying this formula to the example data:

For group A:

$$\begin{aligned} S_A(s,t) := \sum_{i=1}^{n} \tilde{X}_i(s)\tilde{X}_i(t)\,O_i(s)O_i(t)\,\mathbb{1}\{O_i \in A\} \\ &= (1.8-1.93)(2.0-2.17) \\ &\quad + (1.2-1.93)(1.5-2.17) \\ &\quad + (2.8-1.93)(3.0-2.17) \\ &= 1.24. \end{aligned}$$

With $\hat{p}_A(s) = \hat{p}_A(t) = 0.3$ and $n = 10$,

$$\frac{1}{n}\frac{S_A(s,t)}{\hat{p}_A(s)\hat{p}_A(t)} = \frac{1.24}{10 \cdot 0.09} = 1.38.$$

For group B:

$$\begin{aligned} S_B(s,t) &:= \sum_{i=1}^{n} \tilde{X}_i(s)\tilde{X}_i(t)\,O_i(s)O_i(t)\,\mathbb{1}\{O_i \in B\} \\ &= (1.0-1.57)(1.2-1.80) \\ &\quad + (2.2-1.57)(2.4-1.80) \\ &\quad + (1.5-1.57)(1.8-1.80) \\ &= 0.71. \end{aligned}$$

With $\hat{p}_B(s) = \hat{p}_B(t) = 0.3$,

$$\frac{1}{n} \frac{S_B(s,t)}{\hat{p}_B(s)\hat{p}_B(t)} = \frac{0.71}{10 \cdot 0.09} = 0.79.$$

Combining the two group contributions gives the final estimate

$$\hat{k}(s,t) = 1.38 + 0.79 = 2.17.$$

This example shows that only observations available at both time points contribute to the covariance estimation, and that the normalization by $\hat{p}_g(s)\hat{p}_g(t)$ corrects for differing observation probabilities across groups A and B. For instance, if a curve is observed at time s but missing at time t, the product $X_i(s)X_i(t)$ cannot be computed, and the respective contribution to $\hat{k}(s,t)$ is undefined. This illustrates why the restriction to a suitable subdomain, as discussed in Section 2.3.2, is crucial to ensure that a sufficient number of curves contribute to each pairwise covariance estimate.

B Electronic appendix

Data, code and figures are provided in electronic form. Code is available on: `https://github.com/lukasstk/Bachelor_FDA.git`

References

Alwan, H., Pruijm, M., Ponte, B., Ackermann, D., Guessous, I., Ehret, G., ... Bochud, M. (2014). Epidemiology of masked and white-coat hypertension: the family-based SKIPOGH study. *PloS one*, *9*(3), e92522. (Place: United States) doi: 10.1371/journal.pone.0092522

Billingsley, P. (1999). Weak convergence in metric spaces. In *Convergence of probability measures* (pp. 7–79). John Wiley & Sons, Ltd. Retrieved from `https://onlinelibrary.wiley.com/doi/abs/10.1002/9780470316962.ch1` (Section: 1 tex.eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1002/9780470316962.ch1)

Bosq, D. (2000). *Linear Processes in Function Spaces* (Vol. 149; P. Bickel et al., Eds.). New York, NY: Springer. Retrieved 2025-10-04, from `http://link.springer.com/10.1007/978-1-4612-1154-9` doi: 10.1007/978-1-4612-1154-9

Chen, M. G. (2023, June). Estimating Sparsely and Irregularly Observed Multivariate Functional Data. *The ITEA Journal of Test and Evaluation*, *44*(3). Retrieved 2025-10-04, from `https://itea.org/journals/volume-44-3/estimating-sparsely-and-irregularly-observed-multivariate-functional-data/`

Corporation, M., & Weston, S. (2022). *doParallel: Foreach parallel adaptor for the 'parallel' package.* Retrieved from `https://CRAN.R-project.org/package=doParallel`

Davison, A. C., & Hinkley, D. V. (1997). Tests. In *Bootstrap methods and their application* (pp. 136–190). Cambridge: Cambridge University Press.

Gaujoux, R. (2025). *doRNG: Generic reproducible parallel backend for 'foreach' loops.* Retrieved from `https://CRAN.R-project.org/package=doRNG`

Gertheiss, J., Rügamer, D., Liew, B. X. W., & Greven, S. (2024). Functional data analysis: An introduction and recent developments. *Biometrical Journal*, *66*(7), e202300363. Retrieved from `https://onlinelibrary.wiley.com/doi/abs/10.1002/bimj.202300363` (tex.eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1002/bimj.202300363) doi: https://doi.org/10.1002/bimj.202300363

Heitjan, D. F., & Basu, S. (1996). Distinguishing "missing at random" and "missing completely at random". *The American Statistician*, *50*(3), 207–213. Retrieved from `https://doi.org/10.1080/00031305.1996.10474381` (Publisher: ASA Website tex.eprint: https://doi.org/10.1080/00031305.1996.10474381) doi: 10.1080/00031305.1996.10474381

Hsing, T., & Eubank, R. (2013). *Theoretical foundations of functional data analysis, with an introduction to linear operators.* (Pages: 1-333) doi: 10.1002/9781118762547

Hörmann, S., & Kokoszka, P. (2010, June). Weakly dependent functional data. *The Annals of Statistics*, *38*(3), 1845–1884. Retrieved 2025-09-25, from `https://projecteuclid.org/journals/annals-of-statistics/volume-38/issue-3/`

Weakly-dependent-functional-data/10.1214/09-AOS768.full (Publisher: Institute of Mathematical Statistics) doi: 10.1214/09-AOS768

Jolliffe, I. T. (2002). Choosing a subset of principal components or variables. In *Principal component analysis* (pp. 111–149). New York, NY: Springer New York. Retrieved from https://doi.org/10.1007/0-387-22440-8_6 doi: 10.1007/0-387-22440-8_6

Koner, S., & Staicu, A.-M. (2023). Second-generation functional data. *Annual Review of Statistics and Its Application*, *10*(Volume 10, 2023), 547–572. Retrieved from https://www.annualreviews.org/content/journals/10.1146/annurev-statistics-032921-033726 (Publisher: Annual Reviews Type: Journal Article) doi: https://doi.org/10.1146/annurev-statistics-032921-033726

Kraus, D. (2015). Components and completion of partially observed functional data. *Journal of the Royal Statistical Society. Series B (Statistical Methodology)*, *77*(4), 777–801. Retrieved 2025-09-24, from http://www.jstor.org/stable/24775309

Kraus, D. (2019). Inferential procedures for partially observed functional data. *Journal of Multivariate Analysis*, *173*, 583–603. Retrieved from https://www.sciencedirect.com/science/article/pii/S0047259X18304950 doi: https://doi.org/10.1016/j.jmva.2019.05.002

Land Steiermark. (2023). *LUIS - online luftgütedaten.* Retrieved 2023-09-28, from https://app.luis.steiermark.at/luft2/suche.php

Lang, M. (2017). checkmate: Fast argument checks for defensive R programming. *The R Journal*, *9*(1), 437–445. doi: 10.32614/RJ-2017-028

Liebl, D., & Rameseder, S. (2019). Partially observed functional data: The case of systematically missing parts. *Computational Statistics & Data Analysis*, *131*, 104–115. Retrieved from https://www.sciencedirect.com/science/article/pii/S0167947318301944 doi: https://doi.org/10.1016/j.csda.2018.08.011

Little, R. J. A. (1988). A test of missing completely at random for multivariate data with missing values. *Journal of the American Statistical Association*, *83*(404), 1198–1202. Retrieved 2025-09-10, from http://www.jstor.org/stable/2290157 (Publisher: [American Statistical Association, Taylor & Francis, Ltd.])

Little, R. J. A., & Rubin, D. B. (2002). Introduction. In *Statistical analysis with missing data* (pp. 1–23). John Wiley & Sons, Ltd. Retrieved from https://onlinelibrary.wiley.com/doi/abs/10.1002/9781119013563.ch1 (Section: 1 tex.eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1002/9781119013563.ch1) doi: https://doi.org/10.1002/9781119013563.ch1

Microsoft, & Weston, S. (2022). *foreach: Provides foreach looping construct.* Retrieved from https://CRAN.R-project.org/package=foreach

Ofner, M., Hörmann, S., Kraus, D., & Liebl, D. (2025). *Testing the missing completely at random assumption for functional data.* Retrieved from https://arxiv.org/abs/2505.08721 (arXiv: 2505.08721 [stat.ME])

R Core Team. (2023). *R: a language and environment for statistical computing.* Vienna, Austria: R Foundation for Statistical Computing. Retrieved from https://www.R-project.org/

Rubin, D. B. (1976, December). Inference and missing data. *Biometrika*, *63*(3), 581–592. Retrieved from https://doi.org/10.1093/biomet/63.3.581 (tex.eprint: https://academic.oup.com/biomet/article-pdf/63/3/581/756166/63-3-581.pdf) doi: 10.1093/biomet/63.3.581

Scheipl, F., & Goldsmith, J. (2025). *tf: S3 classes and methods for tidy functional data.* Retrieved from https://tidyfun.github.io/tf/

Scheipl, F., Goldsmith, J., & Wrobel, J. (2025). *tidyfun: Tools for tidy functional data.* Retrieved from https://tidyfun.github.io/tidyfun/

Seaman, S., Galati, J., Jackson, D., & Carlin, J. (2013, May). What Is Meant by "Missing at Random"? *Statistical Science*, *28*(2), 257–268. Retrieved 2025-09-14, from https://projecteuclid.org/journals/statistical-science/volume-28/issue-2/What-Is-Meant-by-Missing-at-Random/10.1214/13-STS415.full (Publisher: Institute of Mathematical Statistics) doi: 10.1214/13-STS415

Shang, H. L. (2014, April). A survey of functional principal component analysis. *AStA Advances in Statistical Analysis*, *98*(2), 121–142. Retrieved from https://doi.org/10.1007/s10182-013-0213-1 doi: 10.1007/s10182-013-0213-1

Srivastava, A., & Klassen, E. P. (2016). Functional data and elastic registration. In *Functional and shape data analysis* (pp. 73–123). New York, NY: Springer New York. Retrieved from https://doi.org/10.1007/978-1-4939-4020-2_4 doi: 10.1007/978-1-4939-4020-2_4

Wang, J.-L., Chiou, J.-M., & Müller, H.-G. (2016). Functional data analysis. *Annual Review of Statistics and Its Application*, *3*(Volume 3, 2016), 257–295. Retrieved from https://www.annualreviews.org/content/journals/10.1146/annurev-statistics-041715-033624 doi: https://doi.org/10.1146/annurev-statistics-041715-033624

Yao, F., Müller, H.-G., & Wang, J.-L. (2005). Functional data analysis for sparse longitudinal data. *Journal of the American Statistical Association*, *100*(470), 577–590. Retrieved from https://doi.org/10.1198/016214504000001745 (Publisher: ASA Website tex.eprint: https://doi.org/10.1198/016214504000001745) doi: 10.1198/016214504000001745

Declaration of authorship

I hereby declare that the report submitted is my own unaided work. All direct or indirect sources used are acknowledged as references. I am aware that the Thesis in digital form can be examined for the use of unauthorized aid and in order to determine whether the report as a whole or parts incorporated in it may be deemed as plagiarism. For the comparison of my work with existing sources I agree that it shall be entered in a database where it shall also remain after examination, to enable comparison with future Theses submitted. Further rights of reproduction and usage, however, are not granted here. This paper was not previously presented to another examination board and has not been published.

Location, date

Name